

Seminar On Algebraic Geometry
Lecture Notes 1 (Incomplete version)

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1 Sheaves

Grothendieck's creature in algebraic geometry arises mcuh from differential geometry. Roughly speaking, Grothendieck generalized both differential geometry and algebraic geometry, in order to provide a new geomtric view with algebraic objects.

Definition 1.1. Let $(X, \mathcal{T})$ be a topology space. A **presheaf** $\mathscr{F}$ of category $\mathcal{C}$ is a contravarinat functor

$$\mathscr{F} : \mathsf{Top}(X) \to \mathscr{C},$$

where $\mathrm{Top}(X) = (\mathcal{T}, \subseteq)$ is a preorder category.

In other words, if $\mathscr{F}$ is a presheaf of Ab, then for every open subset U of X, there is an abelian group $\mathscr{F}(U)$ and for every inclusion $U \subseteq V$ of open subsets of X, there is a morphism of abelian groups $\rho_{VU} = \mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$. And it satisfies

(1) $\mathscr{F}(\varnothing) = 0$; (Not necessary)

(2) $\rho_{UU} = \mathrm{id}_U : \mathscr{F}(U) \to \mathscr{F}(U)$;

(3) If $U \subseteq V \subseteq W$ are three open subsets, then $\rho_{WU} = \rho_{WV} \circ \rho_{VU}$.

$$\begin{array}{ccc} U & \longrightarrow & \mathscr{F}(U) \\ \big\downarrow & & \big\uparrow{\scriptstyle \rho_{VU}} \\ V & \longrightarrow & \mathscr{F}(V) \end{array}$$

As a matter of terminology, if : $\mathscr{F}$ is a presheaf on X, we refer to: $\mathscr{F}(U)$ as the **sections** of the presheaf $\mathscr{F}$ over the open set U, and we sometimes use the notation $\Gamma(U, \mathscr{F})$ to denote the group $\mathscr{F}(U)$. We call the maps $\rho_{VU} = \mathrm{res}_{V,U}$ **restriction maps**, and we sometimes write $s \mid_V$ instead of $\rho_{VU}(s)$, if $s \in \mathscr{F}(U)$.

We always assume that $\mathcal{C}$ be Ab, i.e., the category of abelian groups, in order to give the category of presheaves the abelian category constructure.

Simply, a presheaf is just a contravariant functor $\mathscr{F} : \mathcal{D} \to \mathcal{C}$. A sheaf is a presheaf whose sections are, in a technical sense, uniquely determined by their restrictions, via "glueing together".

Definition 1.2. A presheaf $\mathscr{F}$ on a topological space X is a **sheaf** if it satisfies the following supplementary conditions:

(4) **Identity axiom:** If U is an open subset, if $\{V_i\}$ is an open covering of U, and if $s \in \mathscr{F}(U)$ is an element such that $s\mid_{V_i} = 0$ for all i, then $s = 0$. It is equivalanet with that $s\mid_{V_i} = t\mid_{V_i} \implies s = t$.

(5) **Glueing axiom:** If U is an open subset, if $\{V_i\}$ is an open covering of U, and if we have elements $s_i \in \mathscr{F}(V_i)$ for each i, with the property that for each i, j, $s_i\mid_{V_i \cap V_j} = s_j\mid_{V_i \cap V_j}$ then there is an (by (4), unique) element $s \in \mathscr{F}(U)$ such that $s\mid_{V_i} = s_i$ for each i.

Lemma 1.3. Let $\mathscr{F}$ be a sheaf on a topological space X. If U and V are disjoint open subsets of X, then $\mathscr{F}(U \amalg V) \cong \mathscr{F}(U) \times \mathscr{F}(V)$.

Proof. The isomorphism is given by $s \mapsto (s\mid_U, s\mid_V)$, and the inverse is given by $(s_1, s_2) \mapsto s$, where s is glued by s_1 and s_2, since $\{U, V\}$ is naturally an open cover of $U \amalg V$. □

Example 1.4. [Ha, Ex. II 1.1] Let A be an abelian group (resp., ring), and define the **constant presheaf** $\underline{A}_{\text{pre}}$ associated to A on the topological space X to be the presheaf $U \mapsto A$ for all $U \neq \varnothing$, with restriction maps the identity. In particular, the constant presheaf $\underline{\mathbb{Z}}_{\text{pre}}$ associated to $\mathbb{Z}$ is given by $U \mapsto \mathbb{Z}$, and any section $s \in \underline{\mathbb{Z}}_{\text{pre}}(U)$ is an integer $s \in \mathbb{Z}$.

[Ha, Example 1.0.3] Let X be a topological space, and A an abelian group (resp., ring). We define the **constant sheaf** $\underline{A}$ on X determined by A as follows.

Give A the discrete topology, and for any open set $U \in X$, let $\underline{A}(U)$ be the group (resp., ring) of all continuous maps of U into A:

$$\underline{A}(U) := C(U, A).$$

The restriction maps are the usual restrictions of maps. Since A is a discrete topological group (resp., ring), each $\underline{A}(U)$ is endowed with group (resp., ring) structure. Immediately, $\underline{A}$ is an abelian group sheaf (resp., ring) on X.

Note that for every connected open set U, $\underline{A}(U) \cong A$, whence the name "constant sheaf". To see this, let $s \in \underline{A}(U)$ be a section, then $U = \bigcup_{a \in A} s^{-1}(a)$. Since U is connected, there is only one $s^{-1}(a) = U$ and the others are empty. Thus, we obtain a surjective homomorphism below

$$\underline{A}(U) \to A \quad \text{with} \quad s \mapsto s(x), \forall x \in U.$$

It is also injective, since $s(U) = \{0\}$ if and only if $s = 0$.

Therefore, if U is an open set whose connected componentes are open, then $\underline{A}(U) = \prod A$, one for each connected components of U.

Here we define sheaves of modules in order to take a more geometric viewpoint on sheaves. They are studied in [Ha, p. II.5].

Definition 1.5. A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ on X.

Definition 1.6. Let $(X, \mathscr{O}_X)$ be a ringed space. A **sheaf of $\mathscr{O}_X$-modules** is a sheaf $\mathscr{F}$ on X, such that for each open set $U \subseteq X$, the abelian group $\mathscr{F}(U)$ is an $\mathscr{O}_X$-module, and for for each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathscr{F}(U) \to \mathscr{F}(V)$ is compatible with the module structure via the ring homomorphism $\mathscr{O}_X(U) \to \mathscr{O}_X(V)$, that is, for any $a \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$, we have $(a \cdot s) \mid_V = (a \mid_V) \cdot (s \mid_V)$.

Far from being abstract nonsense, sheaf theory generalizes and unifies many geometric objects. Here are some examples from geometry and analysis.

Example 1.7. Let X be a topological space. For each open subset $U \subseteq X$, let $C(U)$ denote the $\mathbb{R}$-algebra of continuous functions $U \to \mathbb{R}$, with restriction maps given by restriction of functions. Then $\mathscr{C}$, defined by $\mathscr{C}(U) = C(U)$, is a sheaf of $\mathbb{R}$-algebras on X; the glueing axiom follows from the local nature of continuity. In particular, if M is a smooth manifold, then the structure sheaf $\mathscr{O}_M$ is the sheaf of smooth functions, given by

$$\mathscr{O}_M(U) = C^\infty(U).$$

Example 1.8. Let X be a complex manifold. For each open subset $U \subseteq X$, let $\mathcal{O}(U)$ denote the $\mathbb{C}$-algebra of holomorphic functions $U \to \mathbb{C}$, with restriction maps given by restriction of functions. Then the structure sheaf $\mathscr{O}_X$, defined by $\mathscr{O}_X(U) = \mathcal{O}(U)$, is a sheaf of $\mathbb{C}$-algebras on X: if $\{U_i\}$ is an open cover of U and $f_i \in \mathcal{O}(U_i)$ agree on all overlaps $U_i \cap U_j$, then there is a unique $f \in \mathcal{O}(U)$ restricting to each f_i, since holomorphicity is a local property. This is the holomorphic analogue of the sheaf of smooth functions on a smooth manifold.

Already here one sees why one works with sheaves rather than only global

sections: on a compact connected complex manifold X, every global holomorphic function is constant, so $\mathscr{O}_X(X) \cong \mathbb{C}$, whereas the sheaf $\mathscr{O}_X$ still encodes rich local holomorphic data.

Let M be a smooth manifold. For each open subset $U \subseteq M$, let $\Omega^k(U)$ denote the $C^\infty(U)$-module of smooth k-forms on U, with restriction maps given by restriction of forms. Then Ω^k is a sheaf of C^∞_M-modules on M.

Crucially, Ω^k is not merely a sheaf of abelian groups: for $f \in C^\infty_M(U)$ and $\omega \in \Omega^k(U)$, the product $f\omega$ is defined pointwise by $(f\omega)_p = f(p)\omega_p$, and restriction is compatible with this structure, making Ω^k a sheaf of modules over the structure sheaf $\mathscr{O}_M$. In particular, $\Omega^0 = \mathscr{O}_M$, and Ω^1 is the sheaf of sections of the cotangent bundle T^*M.

Indeed, sheaves of modules provide an analogue of fibre bundles in algebraic geometry. For example, Ω^k is a $\mathscr{O}_M$-module, and is locally free of rank $\binom{n}{k}$ over $\mathscr{O}_M$, hence corresponds to the vector bundle $\bigwedge^k T^*M$, i.e., $\Omega^k = (-, \bigwedge^k T^*M)$.

There is a deep relation between sheaves of modules with bundles. For example, in differential geometry, the Swan-Serre theorem states that: for X a smooth manifold with $\mathbb{R}$-algebra of smooth functions $C^\infty(X)$, there is an equivalence of categories between that of finite rank smooth vector bundles over X and finitely generated projective modules over $C^\infty(X)$. See [Ne, Chapter 12] for more details.

Example 1.9. Let $p : Y \to X$ be a continuous map. For each open subset $U \subseteq X$, define

$$\Gamma(Y/X)(U) = \{s : U \to Y \mid p \circ s = \mathrm{id}_U,\ s \text{ continuous}\},$$

with restriction maps given by $s \mapsto s|_V$. Then $\Gamma(Y/X)$ is a sheaf on X, called the *sheaf of sections* of p.

If $p : Y \to X$ is a vector bundle, then $\Gamma(Y/X)(U)$ is a $C^\infty(U)$-module, and $\Gamma(Y/X)$ is a sheaf of C^∞_X-modules, locally free of rank equal to the rank of the bundle. Indeed, the name "section" comes from such sections in differential geometry.

Example 1.10. Let M be a smooth manifold, for any open subset $U \subseteq X$, define $C^\infty_c(U)$ be the $\mathbb{R}$-vector space of smooth functions from U to $\mathbb{R}$ with compact support. Then C_c forms a presheaf, which is not a sheaf in general.

For example, let $M = \mathbb{R}$, φ be a bump function such that $\mathrm{supp}(\varphi) =$

$[-1/3, 1/3]$, and $\varphi(0) = 1$. Let $f(x) = \sum_{n\in\mathbb{Z}} \varphi(x-n)$. Then $\mathrm{f}(\varphi) = \bigcup_{n\in\mathbb{Z}}[n-1/3, n+1/3]$, hence $f \notin C_c(\mathbb{R})$. Now let $U_n := (n-1/2, n+2/3)$ be some open subsets covering $\mathbb{R}$. In each U_n, $\mathrm{supp}(f\mid_{U_n}) = [n-1/3, n+1/3]$ which is compact. However, the global section glueing by $f\mid_{U_n}$ is identically f, which is not in $C_c^\infty(\mathbb{R})$.

An interesting result is, as the dual topological space of $C_c^\infty(U)$, the space of distributions $\mathcal{D}'(U)$ forms a sheaf. For a brief introduction to this, see [WkDi, p. 4.5].

Now we come back to classical algebraic geometry. In [Ha, p. I.3], we define the ring of regular functions on an algebraic variety V, denoted by $\mathscr{O}(V)$.

Example 1.11. [Ha, p. II.1.0.1] Let V be an algebraic variety over the field k. For each open set $U \subseteq V$, let $\mathscr{O}_V(U) := \mathscr{O}(U)$ be the ring of regular functions from U to k. Then $\mathscr{O}_V$ is a presheaf of rings on X.

$\mathscr{O}_V$ is a sheaf, which is called the *sheaf of regular functions* on X. This is because the regularity of a function is defined locally.

Now we define the morphisms between presheaves on the same space.

Definition 1.12. For a presheaf $\mathscr{F}$ is a contravariant functor $\mathscr{F} : \mathsf{Top}(X) \to \mathsf{Ab}$, it forms a functor category $[\mathsf{Top}(X)^{\mathsf{op}}, \mathsf{Ab}]$. We define the category of presheaves as $\mathsf{PSh}(X) = [\mathsf{Top}(X)^{\mathrm{op}}, \mathsf{Ab}]$. In general, we define $\mathsf{PSh}(X, \mathcal{C}) = [\mathsf{Top}(X)^{\mathrm{op}}, \mathcal{C}]$, for any category $\mathcal{C}$.

Thus, the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation, which consists of morphisms of abelian groups $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for each open set U, such that whenever $V \subseteq U$ is an inclusion, the diagram

$$\begin{array}{ccc} \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ \downarrow{\scriptstyle \rho_{UV}} & & \downarrow{\scriptstyle \rho_{UV}} \\ \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

commutes. Or we can simply say that $\varphi(s\mid_V) = \varphi(s)\mid_V$, for any section $s \in \mathscr{F}(U)$.

We can simply define the category of sheaves on X to be the full subcategory $\mathsf{Sh}(X) := \mathsf{Sh}(X, \mathsf{Ab})$ of $\mathsf{PSh}(X)$.

Definition 1.13. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves. We define the presheaf kernel of φ, presheaf cokernel of φ, and presheaf image of φ to be the presheaves given by $U \mapsto \ker \varphi(U)$, $U \mapsto \operatorname{coker} \varphi(U)$, and $U \mapsto \operatorname{im} \varphi(U)$ respectively.

Indeed, PSh is an abelian category. We denote the cokernel of φ in PSh by $\operatorname{coker}_{\mathsf{PSh}} \varphi$ and the image of φ in PSh by $\operatorname{im}_{\mathsf{PSh}} \varphi$, since they are not sheaves in general. However, the kernel of φ in PSh is a sheaf, by the lemma below.

Lemma 1.14. If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves, then the presheaf kernel of φ, $\ker \varphi$ is also a sheaf.

Proof. Notice that $(\ker \varphi)(U) \subseteq \mathscr{F}(U)$ hence we can use the sheaf axiom on $\mathscr{F}$. □

2 Stalks & Germs

Before we investigate the category of sheaves and operations on sheaves, we first introduct the stalks of sheaves.

Definition 2.1 (Stalks)**.** If $\mathscr{F}$ is a presheaf on X, and if P is a point of X, we define the *stalk* $\mathscr{F}_P$ of $\mathscr{F}$ at P to be the direct limit (colimit) of the groups $\mathscr{F}(U)$ for all open sets U containing P, via the restriction maps ρ, as

$$\mathscr{F}_P := \varinjlim \mathscr{F}(U).$$

The direct limit means that, for any open neighborhood U of P, there exists a homomorphism $\rho_U : \mathscr{F}(U) \to \mathscr{F}_P$, such that, for compatible homomorphisms $\lambda_U : \mathscr{F}(U) \to A$, there exists a unique homomorphism $\mathscr{F}_P \to A$.

Elements in $\mathscr{F}_P$ are called *germs* of sections of $\mathscr{F}$ at the point P.

In fact, the index category $\mathsf{Top}_P(X)$ with all open neighborhood U of P is a *filtered category*, since for any open neighborhood U and V of P, $U \cap V$ is also an open neighborhood of P. By [Va, p. 1.4 D], the germ in $\mathscr{F}_P$ has the form of $\langle U, s\rangle$ while U is the open neighborhood of P and $s \in \mathscr{F}(U)$ under the equivalence relation that

$$\langle U, s\rangle = \langle V, t\rangle \iff \exists \text{ an open neighborhood } W \subseteq U \cap V \text{ of } P \text{ s.t. } s\,|_W = t\,|_W\,.$$

In fact, taking stalks gives a functor $\mathsf{Sh}(X) \to \mathsf{Ab}$, which is exact and left-adjoint.

Example 2.2. Let X be a complex manifold and let $\mathscr{O}_X$ be the sheaf of holomorphic functions on X. If $\dim_{\mathbb{C}} X = n$ and $z_1, \ldots, z_n$ are local holomorphic coordinates centered at $P \in X$, then the germ of a holomorphic function is uniquely determined by its Taylor expansion at P. Hence

$$\mathscr{O}_{X,P} \cong \mathbb{C}\{z_1, \ldots, z_n\},$$

the ring of convergent power series in n variables. This stalk is a local ring, whose maximal ideal consists of germs vanishing at P.

Example 2.3. The name of germs came from analytic continuation. See [Ah, Chapter 8].

A *function element* is a pair (D, f) consisting of an open disk $D \subset \mathbb{C}$ and a holomorphic function f on D. Two function elements (D, f) and (E, g) define the same *germ* at $z_0 \in D \cap E$ if $f = g$ in a neighborhood of z_0. The set of all such germs, denoted $\mathcal{O}$, carries a natural topology: for each function element (D, f), the collection $\{[f]_z : z \in D\}$ is a basic open neighborhood of each of its germs. The projection

$$\pi : \mathcal{O} \to \mathbb{C}, \qquad [f]_z \mapsto z,$$

is then a local homeomorphism, because the map $z \mapsto [f]_z$ is a homeomorphism from D onto $\{[f]_z : z \in D\}$ with inverse π. Thus $\mathcal{O}$ is a one-dimensional complex manifold, i.e. a Riemann surface, and π is holomorphic. Its connected components are the Riemann surfaces of the *global analytic functions* obtained by analytic continuation: continuing a germ $[f]_a$ along a curve γ from a to b amounts to lifting γ to $\mathcal{O}$, so the continuation becomes a single-valued holomorphic function on the component containing $[f]_a$.

Example 2.4. More generally, let X be a topological space. An *étale space* over X is a continuous map $\pi : \mathcal{F} \to X$ which is a local homeomorphism. The fiber $\mathcal{F}_x = \pi^{-1}(x)$ is the *stalk* at x. If each stalk carries the structure of an abelian group and the group operations are continuous, this is the old-fashioned definition of a *sheaf of abelian groups* on X; the sections over an open set U are the continuous maps $s : U \to \mathcal{F}$ with $\pi \circ s = \mathrm{id}_U$. The modern sheaf $\mathscr{F}$ associated with an étale space is obtained by taking all sections over open subsets.

3 Local Nature of Sheaves I

Lemma 3.1. For every $P \in U$, there is a map $\mathscr{F}(U) \to \mathscr{F}_P$, sending a section $s \in \mathscr{F}(U)$ to the equivalence class of (U, s), denoted s_P.

If $\mathscr{F}$ is a sheaf, a section $s \in \mathscr{F}(U)$ of $\mathscr{F}$ over U is determined by its images in the stalks $\mathscr{F}_x$ for all $x \in U$, i.e. $s = 0$ if and only if $s_P = 0$, for all $P \in U$. Equivalently, $s = t \in \mathscr{F}(U)$ if and only if $s_P = t_P \in \mathscr{F}_P$, for all $P \in U$.

Proof. For any germ (s_P, U_P), such that $S_P = 0$, we can obtain the zero section by glueing s_P, since U_P forms an open cover of X. By the identity axiom, any section glueing by s_P must be zero. □

Proposition 3.2. [Va, 2.4 N], [Ha, Ex 1.3 (a)] Consider the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves on the topology space X. Then, the conditions below are equivalent:

(1) φ is a monomorphism.

(2) φ is injective on the level of stalks: $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is injective for all $P \in X$.

(3) $\ker \varphi = 0$.

(4) φ is injective on the level of open sets : $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is a monomorphism of Ab, thus injective and $\ker \varphi(U) = (0)$, for all open set $U \subseteq X$.

We say that φ is *injective* if it satisfies the above equivalent conditions.

Proof. We prove this by (2) $\Longrightarrow$ (1) $\Longrightarrow$ (3) $\Longrightarrow$ (4) $\Longrightarrow$ (2).

(2) $\Longrightarrow$ (1) : By 3.1.

(1) $\Longrightarrow$ (3) : By the definition of $\ker \varphi$ we could obtain an inclusion morphism $i : \ker \varphi \to \mathscr{F}$ and a zero morphism $0 : \ker \varphi \to \mathscr{F}$ sending $(\ker \varphi)(U) \mapsto (0)$ inducing the commutative diagram below

$$\ker \varphi \underset{0}{\overset{i}{\rightrightarrows}} \mathscr{F} \xrightarrow{\varphi} \mathscr{G}.$$

Since φ is a monomorphism, i must be 0, which means that $\ker \varphi = 0$.

(3) $\Longrightarrow$ (4) : Notice that $\ker(\varphi(U)) = (\ker \varphi)(U)$.

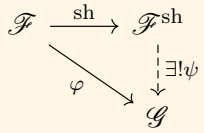
(4) $\Longrightarrow$ (2) : If $\ker \varphi(U) = (0)$ for all open sets $U \subseteq X$, then for any $P \in X$, and germs $s_P \in \ker \varphi_P$, there exists an open neighborhood U of P, such that $s_P = \langle U, s \rangle$ for some $s \in \ker \varphi(U)$. By the assumption, $s = 0$, hence $s_P = 0$.

□

Since $\mathrm{im}_{\mathsf{PSh}}$ is not a sheaf in general, the case of epimorphisms is more complicated. We need to introduce the sheafification of presheaves, in order to obtain the image of a morphism of sheaves.

4 Sheafification

Proposition 4.1. [Ha, Proposition 1.2] Given a presheaf $\mathscr{F}$, there is a sheaf $\mathscr{F}^{\mathrm{sh}}$ and a morphism $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ with the following universal property: for any sheaf $\mathscr{G}$ and any morphism of presheaves $\varphi : \mathscr{F} \to \mathscr{G}$ (where $\mathscr{G}$ is regarded as a presheaf), there is a unique morphism of sheaves $\psi : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ such that $\varphi = \psi \circ \mathrm{sh}$.



Furthermore the pair $(\mathscr{F}^{\mathrm{sh}}, \mathrm{sh})$ is unique up to a unique isomorphism. We call $\mathscr{F}^{\mathrm{sh}}$ *the sheaf associated to the presheaf* $\mathscr{F}$, or *the sheafification of* $\mathscr{F}$.

Proof. We construct the sheafification $\mathscr{F}^{\mathrm{sh}}$ as follows. For any open subset U, let $\mathscr{F}^{\mathrm{sh}}(U)$ be the set of functions

$$s : U \longrightarrow \coprod_{P \in U} \mathscr{F}_P$$

such that

(1) for each $P \in U$, $s(P) \in \mathscr{F}_P$, and

(2) for each $P \in U$, there is an open neighborhood $V \subseteq U$ of P, and an element $t \in \mathscr{F}(V)$, such that for all $Q \in V$, the germ t_Q of t at Q is equal to $s(Q)$.

By construction, sh is naturally defined by sending $s \in \mathscr{F}(U)$ to the function $P \mapsto s_P$. The restriction maps are defined by restricting the domain of these functions.

We now verify that $\mathscr{F}^{\mathrm{sh}}$ is a sheaf. For the identity axiom, let $U \subseteq X$ be open, let $\{U_i\}$ be an open cover of U, and let $s \in \mathscr{F}^{\mathrm{sh}}(U)$ satisfy $s|_{U_i} = 0$ for all i. Since the restriction maps are restrictions of functions, $s|_{U_i}(P) = s(P)$ for every $P \in U_i$. Given $P \in U$, choose i with $P \in U_i$. Then

$$s(P) = s|_{U_i}(P) = 0.$$

Hence $s = 0$.

For the glueing axiom, suppose we are given sections $s_i \in \mathscr{F}^{\mathrm{sh}}(U_i)$ such that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j. Define $s : U \to \coprod_{P \in U} \mathscr{F}_P$ by $s(P) = s_i(P)$ whenever $P \in U_i$. This is well defined by the compatibility on overlaps. To see that $s \in \mathscr{F}^{\mathrm{sh}}(U)$, fix $P \in U$ and choose i with $P \in U_i$. Since $s_i \in \mathscr{F}^{\mathrm{sh}}(U_i)$, there is an open neighborhood $V \subseteq U_i$ of P and an element $t \in \mathscr{F}(V)$ such that $t_Q = s_i(Q)$ for all $Q \in V$. Then $t_Q = s(Q)$ for all $Q \in V$, so the local condition holds at P. Thus $s \in \mathscr{F}^{\mathrm{sh}}(U)$, and $s|_{U_i} = s_i$ for every i, because the two functions agree at each point of U_i. Therefore $\mathscr{F}^{\mathrm{sh}}$ is a sheaf.

Now we prove the universal property. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves, with $\mathscr{G}$ a sheaf. Let $s \in \mathscr{F}^{\mathrm{sh}}(U)$. For each $P \in U$, by the construction of $\mathscr{F}^{\mathrm{sh}}$, there is an open neighborhood $V_P \subseteq U$ of P and an element $t^P \in \mathscr{F}(V_P)$ such that

$$s(Q) = t^P_Q \qquad \text{for all } Q \in V_P.$$

We claim that for any $P, Q \in U$, the sections $\varphi(t^P)$ and $\varphi(t^Q)$ agree on $V_P \cap V_Q$. Indeed, for every $R \in V_P \cap V_Q$, we have

$$\varphi_R\big(t^P_R\big) = \varphi_R\big(s(R)\big) = \varphi_R\big(t^Q_R\big).$$

Thus the germs of $\varphi(t^P)$ and $\varphi(t^Q)$ at every point of $V_P \cap V_Q$ coincide. Since $\mathscr{G}$ is a sheaf, Lemma 3.1 implies

$$\varphi(t^P)|_{V_P \cap V_Q} = \varphi(t^Q)|_{V_P \cap V_Q}.$$

Therefore, by the glueing axiom for the sheaf $\mathscr{G}$, there exists a unique section $\psi(s) \in \mathscr{G}(U)$ such that

$$\psi(s)|_{V_P} = \varphi(t^P) \qquad \text{for all } P \in U.$$

The section $\psi(s)$ is independent of the choices of V_P and t^P: if W_P, u^P is another choice, then on $V_P \cap W_P$ the sections $\varphi(t^P)$ and $\varphi(u^P)$ have the same germs at every point, hence are equal because $\mathscr{G}$ is a sheaf; uniqueness in the glueing axiom then gives the same global section.

It is immediate from the local description that ψ is compatible with restriction maps, hence ψ is a morphism of sheaves. Moreover, for $t \in \mathscr{F}(U)$, the section $\psi(\mathrm{sh}(t))$ agrees locally with $\varphi(t)$, so $\psi \circ \mathrm{sh} = \varphi$.

Finally, suppose that $\psi' : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ is another morphism of sheaves such that

$\psi' \circ \mathrm{sh} = \varphi$. For $s \in \mathscr{F}^{\mathrm{sh}}(U)$ and $P \in U$, using the same local data V_P, t^P, we have

$$\psi'(s)|_{V_P} = \psi'(\mathrm{sh}(t^P))|_{V_P} = \varphi(t^P) = \psi(s)|_{V_P}.$$

Since $\mathscr{G}$ is a sheaf, this implies $\psi'(s) = \psi(s)$. Hence $\psi' = \psi$. The uniqueness of $(\mathscr{F}^{\mathrm{sh}}, \mathrm{sh})$ up to a unique isomorphism follows formally from the universal property. □

Lemma 4.2. [Va, p. 2.4 M] The sheafification $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ induces an isomorphism on stalks:

$$\mathrm{sh}_P : \mathscr{F}_P \longrightarrow \mathscr{F}^{\mathrm{sh}}_P, \qquad \forall P \in X.$$

Conversely, let $\mathscr{F}$ be a presheaf, let $\mathscr{G}$ be a sheaf, and let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves. If the induced map $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is an isomorphism for all $P \in X$, then the induced morphism $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ is an isomorphism.

Proof. First we prove that sh_P is injective. Let $\alpha \in \mathscr{F}_P$, and choose a representative $s \in \mathscr{F}(U)$ of α. Suppose that $\mathrm{sh}_P(\alpha) = 0$. Then $\mathrm{sh}(s)_P = 0$ in $\mathscr{F}^{\mathrm{sh}}_P$. Hence there is an open neighborhood $V \subseteq U$ of P such that

$$\mathrm{sh}(s)|_V = 0 \qquad \text{in } \mathscr{F}^{\mathrm{sh}}(V).$$

For every $Q \in V$, we have

$$\mathrm{sh}(s)(Q) = s_Q.$$

Therefore $s_Q = 0$ for every $Q \in V$. In particular, taking $Q = P$, we obtain $s_P = 0$ in $\mathscr{F}_P$, that is, $\alpha = 0$. Thus sh_P is injective.

Next we prove that sh_P is surjective. Let $\beta \in \mathscr{F}^{\mathrm{sh}}_P$, and choose a representative $\sigma \in \mathscr{F}^{\mathrm{sh}}(U)$ of β. By the construction of $\mathscr{F}^{\mathrm{sh}}$, after possibly shrinking U to an open neighborhood V of P, there exists $t \in \mathscr{F}(V)$ such that

$$t_Q = \sigma(Q) \qquad \text{for all } Q \in V.$$

Hence $\mathrm{sh}(t)|_V = \sigma|_V$, and therefore

$$\mathrm{sh}_P(t_P) = \mathrm{sh}(t)_P = \sigma_P = \beta.$$

Thus sh_P is surjective. Consequently sh_P is an isomorphism for every $P \in X$.

For the converse, the universal property of sheafification gives a unique morphism $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ such that $\varphi^{\mathrm{sh}} \circ \mathrm{sh} = \varphi$. On stalks, since sh_P is an isomorphism, we

obtain

$$\varphi_P^{\mathrm{sh}} = \varphi_P \circ \mathrm{sh}_P^{-1}.$$

By assumption φ_P is an isomorphism for every $P \in X$. Hence φ_P^{sh} is an isomorphism for every $P \in X$. By Proposition 5.3, φ^{sh} is an isomorphism. □

Proposition 4.3. [Va, 2.4 H] Let $\mathscr{F}, \mathscr{G} \in \mathrm{PSh}(X)$ be a presheaf of X, let $\mathscr{F}^{\mathrm{sh}}, \mathscr{G}^{\mathrm{sh}} \in \mathrm{Sh}(X)$ be the sheafification of them. If there is a morphism $\varphi : \mathscr{F} \to \mathscr{G}$, then there exists a (unique) natural induced morphism of sheaves $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}^{\mathrm{sh}}$, such that the following diagram commutes.

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\varphi} & \mathscr{G} \\ \downarrow{\scriptstyle \mathrm{sh}} & & \downarrow{\scriptstyle \mathrm{sh}} \\ \mathscr{F}^{\mathrm{sh}} & \xrightarrow{\varphi^{\mathrm{sh}}} & \mathscr{G}^{\mathrm{sh}} \end{array}$$

Proof. This follows from the universal property of sheafification. □

Remark 4.4. Indeed, Proposition 4.3 shows that the sheafification is a functor $\mathrm{sh} : \mathsf{PSh} \to \mathsf{Sh}$. Moreover, it is left-adjoint to the inclusion functor $\mathsf{Sh} \to \mathsf{PSh}$, i.e., for any presheaf $\mathscr{F}$ and sheaf $\mathscr{G}$,

$$\mathrm{Hom}_{\mathsf{Sh}}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}) \cong \mathrm{Hom}_{\mathsf{PSh}}(\mathscr{F}, \mathscr{G}).$$

Example 4.5. Let A be an abelian group, and X be a topological space. We have defined the constant presheaf $\underline{A}_{\mathrm{pre}}$ on X, and the constant sheaf $\underline{A}$ on X in Example 1.4. We claim that $\underline{A}$ is the sheafification of $\underline{A}_{\mathrm{pre}}$.

Indeed, the natural morphism $\underline{A}_{\mathrm{pre}} \to \underline{A}$ is the identity on stalks: both stalks are A, since a germ of a locally constant function is constant on a small neighborhood. Hence this morphism is an isomorphism on all stalks, and therefore the induced morphism $\underline{A}_{\mathrm{pre}}^{\mathrm{sh}} \to \underline{A}$ is an isomorphism on stalks by Lemma 4.2. It follows that $\underline{A}$ is the sheafification of $\underline{A}_{\mathrm{pre}}$.

One can check that each sheaves of abelian groups on X is a $\underline{\mathbb{Z}}$-sheaves on X by definition. We will prove this by $\mathscr{H}om$ in the next time.

Now we can define the cokernel and image of a morphism of sheaves, hence giving a complete description of the abelian category of sheaves.

Definition 4.6. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves.

- We define the kernel of φ to be the presheaf kernel of φ, $\ker \varphi$.

- We define the cokernel of φ to be the sheafification of the presheaf cokernel of φ, $\operatorname{coker}\varphi := (\operatorname{coker}_{\mathsf{PSh}}\varphi)^{\mathrm{sh}}$.
- We define the image of φ to be the sheafification of the presheaf image of φ, $\operatorname{im}\varphi := (\operatorname{im}_{\mathsf{PSh}}\varphi)^{\mathrm{sh}}$.

Corollary 4.7. [Ha, Ex 1.4] Let $\varphi : \mathscr{F} \to \mathscr{G}$ be an injective morphism of sheaves. Then the induced map $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}^{\mathrm{sh}}$ of sheafifications is injective. Therefore, we can view $\operatorname{im}\varphi$ as a natural subsheaf of $\mathscr{G}$.

Proof. We check this on the level of stalks. Notice that $\varphi_P^{\mathrm{sh}} = \mathrm{sh}_P \circ \varphi_P \circ \mathrm{sh}_P^{-1}$ since sh_P is an isomorphism, by 4.2. Therefore, φ_P^{sh} is injective for all $P \in X$, since φ_P is injective for all $P \in X$. □

5 Local Nature of Sheaves II

By giving the defintion of image in the category of sheaves, we can now give an analogue of Proposition 3.2 for epimorphisms.

Proposition 5.1. [Va, 2.4 O], [Ha, Ex 1.3] Consider the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves on the topology space X. Then, the conditions below are equivalent:

(1) φ is an epimorphism.

(2) φ is surjective on the level of stalks: $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is surjective for all $P \in X$.

(3) $\operatorname{im}\varphi = \mathscr{G}$.

(4) For every open set $U \subseteq X$, and for every section $s \in \mathscr{G}(U)$, there is a covering $\{U_i\}$ of U, and sections $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s\,|_{U_i}$ for all i.

We say that φ is *surjective* if it satisfies the above equivalent conditions.

Proof. We prove this by (2) $\implies$ (1) $\implies$ (3) $\implies$ (4) $\implies$ (2). The proof of (2) $\implies$ (1) and (1) $\implies$ (3) is similiar to Proposition 3.2.

(3) $\implies$ (4) : Let U be an open set, and $s \in \mathscr{G}(U)$ be a section. Since $\operatorname{im}\varphi = \mathscr{G}$, for each $P \in U$, there exists an open neighborhood $V_P \subseteq U$ of P, and a section $t^P \in \mathscr{F}(V_P)$ such that $\varphi(t^P) = s\,|_{V_P}$, because of the construction of the sheafification. Then $\{V_P\}_{P\in U}$ is an open cover of U, and the sections t^P satisfy $\varphi(t^P) = s\,|_{V_P}$ for all $P \in U$.

(4) $\implies$ (2) : For every point $P \in X$, and every germ $\langle s, U\rangle \in \mathscr{G}_P$, there is a covering $\{U_i\}$ of U and sections $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s\mid_{U_i}$, by assumption. Choose an open neighborhood U_i of P, then $\langle s, U\rangle = \langle s, U_i\rangle = \langle \varphi(t_i), U_i\rangle$ in $\mathscr{G}_P$. This proves φ_P is surjective. □

Corollary 5.2. [Ha, Ex 1.2] A sequence of sheaves

$$\cdots \to \mathscr{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathscr{F}^{i} \xrightarrow{\varphi^{i}} \mathscr{F}^{i+1} \to \cdots$$

is exact if and only if the induced sequence of stalks

$$\cdots \to \mathscr{F}_P^{i-1} \xrightarrow{\varphi_P^{i-1}} \mathscr{F}_P^{i} \xrightarrow{\varphi_P^{i}} \mathscr{F}_P^{i+1} \to \cdots$$

for all $P \in X$.

Proposition 5.3. [Ha, Proposition 1.1] Let Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves on a topological space X. Then φ is an isomorphism if and only if the induced map on the stalk $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is an isomorphism for every $P \in X$.

Proof. By Proposition 3.2 and Proposition 5.1. □

Proposition 5.4. [Ha, Ex 1.2] For any morphism of sheaves $\varphi : \mathscr{F} \to \mathscr{G}$, for each point $P \in X$, $(\ker \varphi)_P = \ker(\varphi_P)$ and $(\operatorname{im} \varphi)_P = \operatorname{im}(\varphi_P)$.

Proof. Left to readers. □

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